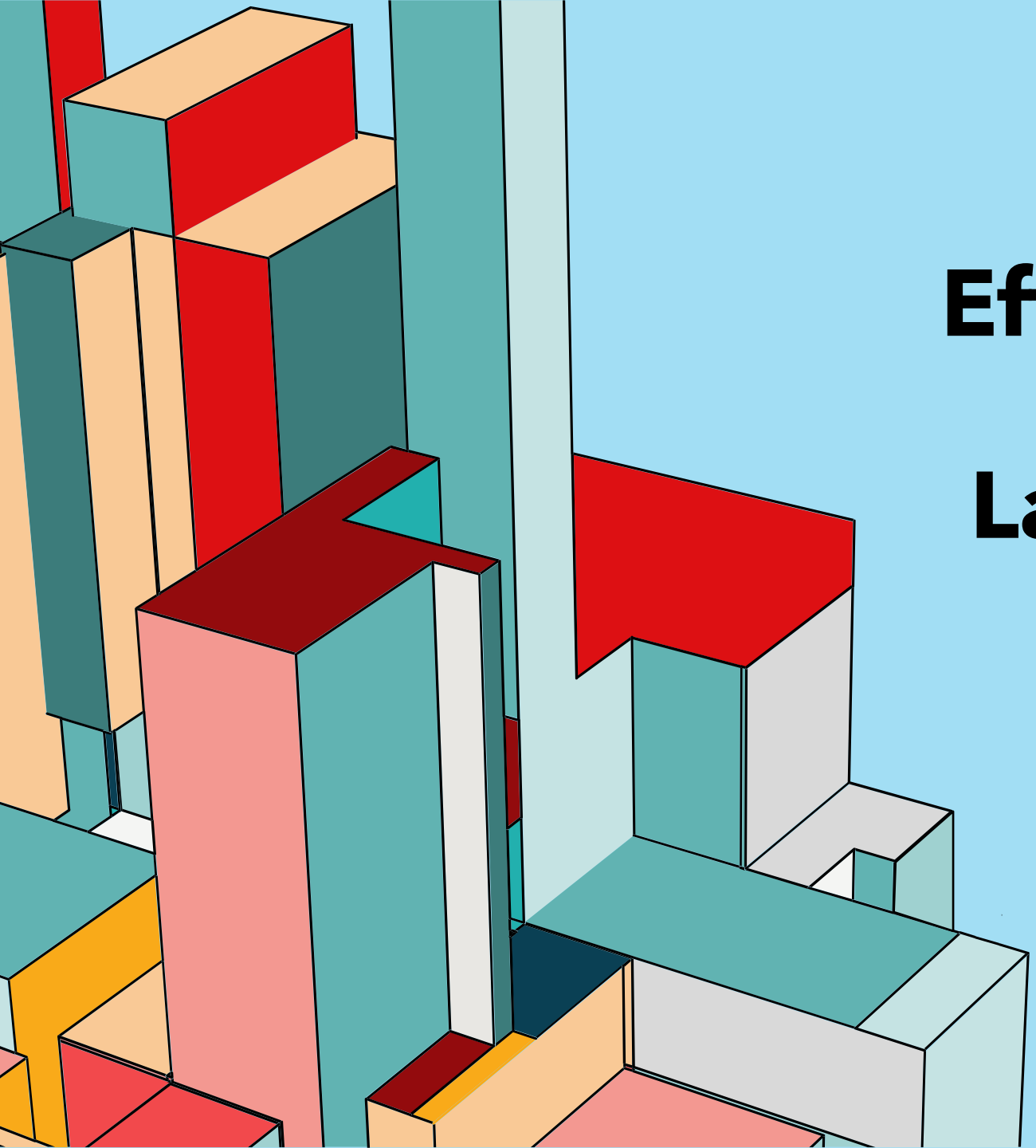


Efficient Adaptation of Multimodal Large Language Models for Visual Storytelling

Politecnico di Torino

FONDAZIONE links PASSION FOR INNOVATION

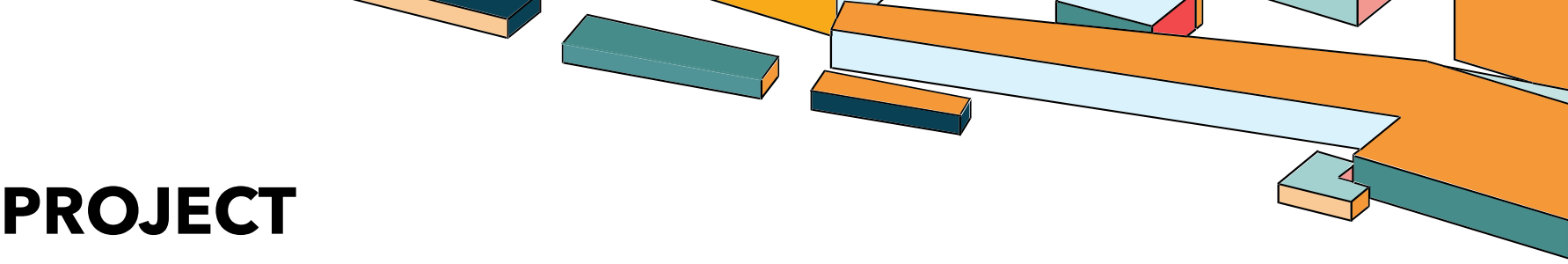
PROJECT PRESENTATION

This project focuses on developing a **Multimodal Large Language Model** (MLLM) system to automatically generate a story from a sequence of images.



AI
(MLLM)

The family got together for a cookout. They had a lot of delicious food. The dog was happy to be there. They had a great time on the beach.



VALUE-DRIVEN PROJECT

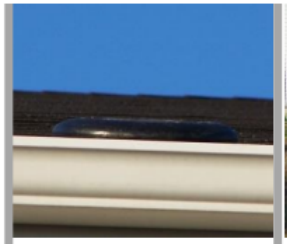
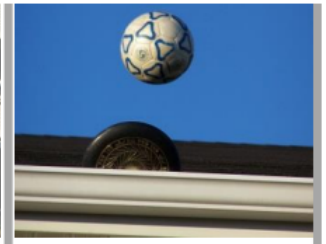
This project enhances **innovation, accessibility, and cultural preservation** by developing an **AI-driven platform that generates personalised, curator-validated narratives** at heritage sites. It deepens visitor engagement at under-visited monuments and widens access to cultural content, creating value through automation, creativity, and digital inclusion.

NARRATIA
LIVE HISTORY

The project is carried out under the supervision of **LINKS Foundation**, within the framework of the European project "**NARRATIA**."

DATA

- VIST
- AESOP
- VWP

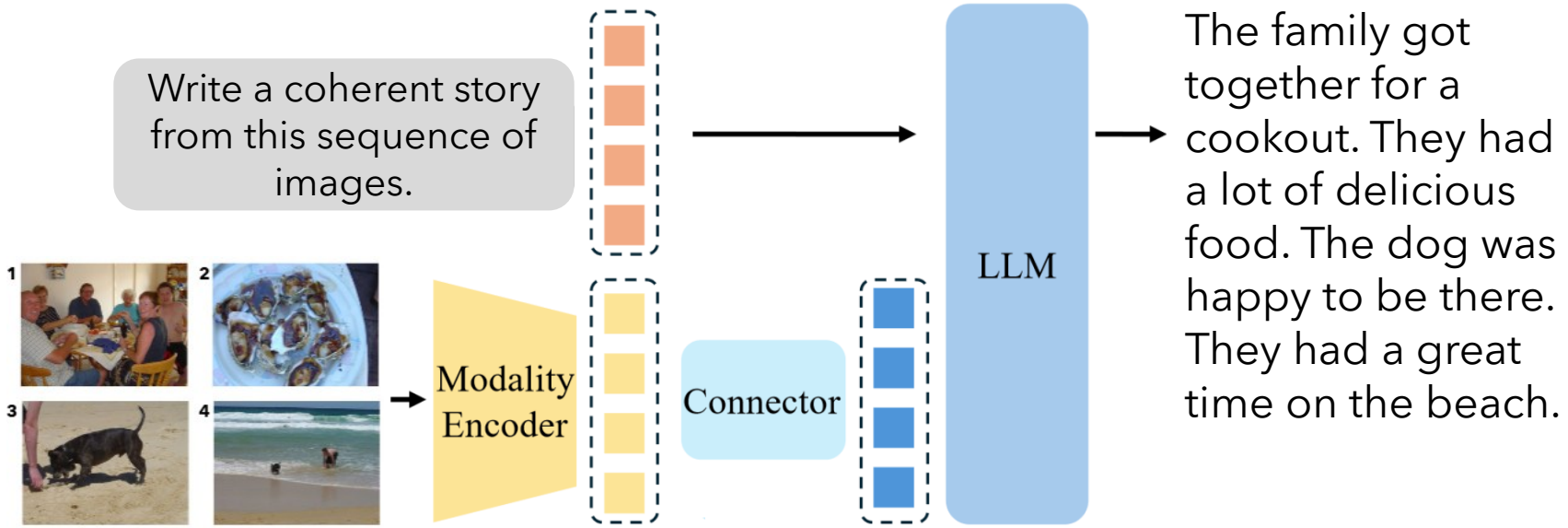
					
DII	A black frisbee is sitting on top of a roof.	A man playing soccer outside of a white house with a red door.	The boy is throwing a soccer ball by the red door.	A soccer ball is over a roof by a frisbee in a rain gutter.	Two balls and a frisbee are on top of a roof.
DIS	A roof top with a black frisbee laying on the top of the edge of it.	A man is standing in the grass in front of the house kicking a soccer ball.	A man is in the front of the house throwing a soccer ball up	A blue and white soccer ball and black Frisbee are on the edge of the roof top.	Two soccer balls and a Frisbee are sitting on top of the roof top.
SIS	A discus got stuck up on the roof.	Why not try getting it down with a soccer ball?	Up the soccer ball goes.	It didn't work so we tried a volley ball.	Now the discus, soccer ball, and volleyball are all stuck on the roof.

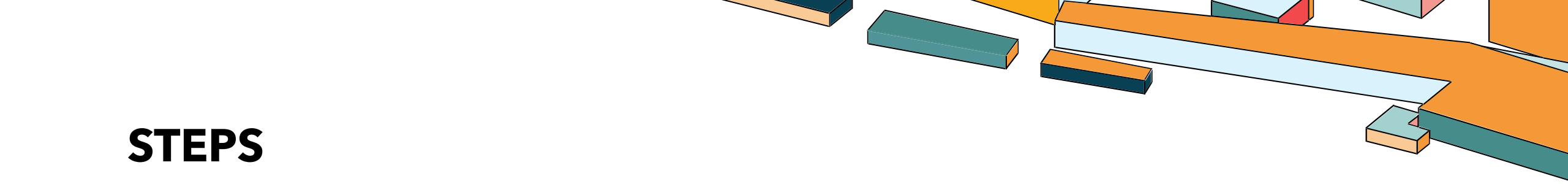
Descriptions of Images in Isolation (DII); Descriptions of Images in Sequence (DIS); **Stories of Images in Sequence (SIS)**.

TASK

Visual Storytelling (VS): generating a coherent narrative from a sequence of images by combining visual understanding with natural language generation.

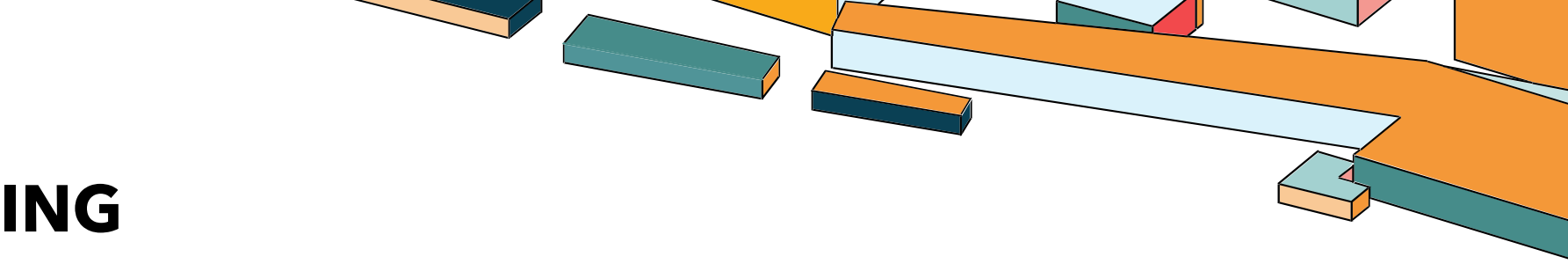
The main goal of this project is to fine-tune a **Multimodal Large Language Model (MLLM)** using **Parameter-Efficient Fine-Tuning (PEFT)** techniques to improve its performance on domain-specific visual storytelling.





STEPS

- Data Exploration
- Zero-shot evaluation of the MLLM on the VS task
- Training (adaptation) of the MLLM for the VS task.
- Evaluation on the validation splits

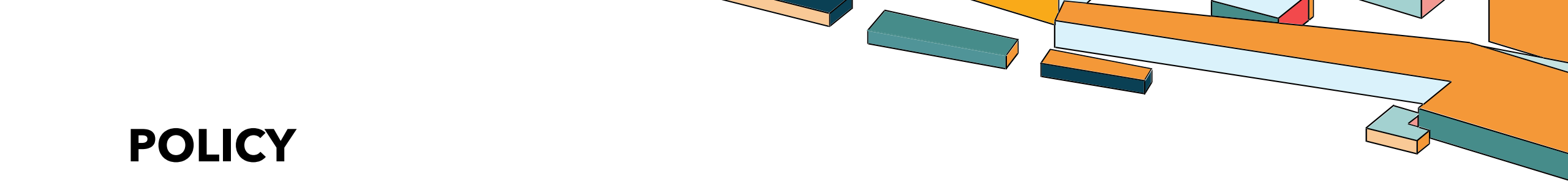


LIGHT MENTORING

- **First (longer) meeting** as an introduction to the existing pipeline and methodology, context and data presentation.

- **Weekly one-hour calls** with students for the whole duration of the semester.

- Feel free to reach out via MS Teams or email at any time for any questions or doubts.



POLICY

- Both project descriptions and implementations will be part of a repository group published on GitHub.

- The repositories will be public unless requests from the organization that will be discussed.

CONTACTS

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